

Muhammad Usman

 uthmanhere |  uthmanhere@gmail.com |  +34 627 226 897 |  Barcelona

SUMMARY

Research engineer specializing in HPC, SmartNIC/DPU offloading, MPI, OpenMP, and RDMA. Strong experience in system-level optimization, compiler/runtime internals, and kernel networking, with peer-reviewed publications and SC-level tutorials.

WORK EXPERIENCE

Research Engineer – Barcelona Supercomputing Center Jun 2022 - Present

- *Enabling BlueField DPU Offloading of MPI Calls* in OpenMPI using the underlying UCX framework. Ported and evaluated on entire OSU Microbenchmark to prove no performance downgrade. Transformed miniWeather application to non-blocking communication and added halo exchange such that a part of compute and node-to-node communication can be offloaded to BlueField DPU to gain 18% speedup using the proposed framework.
[\[SC25 Publication\]](#) [\[HiPEAC info Article\]](#) [\[SC25 Tutorial\]](#) [\[Repo\]](#)
- *BlueField DPU Offloading in LLVM/OpenMP Target Plugin* using DOCA SCC. Ported and evaluated SPEC-accel benchmarks to test framework on BlueField 2 and 3.
[\[LLVM-HPC Publication\]](#) [\[HPCWire Article\]](#) [\[HPCCKP Short Talk\]](#) [\[Repo\]](#)

Software Engineer – ARM (previously DreamBig Semiconductor) Jan 2021 - Jun 2022

- Exploring implementation *GPUDirect RDMA & GPUDirect Async* in RDMA Device Driver Subsystem
- Testing and Documenting steps to reproduce and linux implementation of RoCE with Open vSwitch for *Overlay Networks & Link Aggregation* using TC Rules.
- Porting and Profiling of an FPGA-based TCP Offload Engine to enable RDMA applications as an iWARP implementation.
- Porting and Profiling of an XDMA-based TOE Device Driver to QDMA.

EDUCATION

BS in Electrical Engineering at **National University of Sciences & Technology**

PUBLICATIONS

- M. Usman, M. Benito, S. Iserte, and A. J. Peña, *ODOS-MPI: HPC-Friendly SmartNIC Offloading of Computation/Communication Kernels* in The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC25), St. Louis, MO, USA, Nov. 2025.
- M. Usman, S. Iserte, R. Ferrer, and A. J. Peña, *DPU offloading programming with the OpenMP API*, in The Ninth Workshop on the LLVM Compiler Infrastructure in HPC (LLVM-HPC), Denver, CO, USA, Nov. 2023.
- Z. A. Khan, T. Hussain, U. Zabit, M. Usman, E. Ayguade, *PH-RLS: A parallel hybrid recursive least square algorithm for self-mixing interferometric laser sensor*, IET Optoelectronics, Oct. 2021.

TUTORIAL & TEACHING EXPERIENCE

- BSC Training Course: *Introduction to SYCL Programming*. 2026
- *In-Network Computing with SmartNICs for HPC Applications*. SC24, ISC25, SC25.
- Summer School: *Supercomputing and Parallel Programming*. Namal University, 2024.

TRAININGS

2025	SC25 Tutorial: Advanced MPI Programming
2025	SC25 Tutorial: Core-Level CPU Performance Engineering
2024	BSC Training Course: Introduction to CUDA Programming
2023	BSC Training Course: Advance Performance Analysis & Tools
2022	HOTI Tutorial: Introduction to Network Technologies for HPC and AI
2022	BSC Training Course: Parallel Programming Workshop
2022	PUMPS+AI Summer School

SKILLS

HPC	Open MPI internals, LLVM/OpenMP Target Offloading internals, UCX, CUDA, DOCA, NCCL, Fortran, Extrae, Paraver
Linux Kernel	Network Driver Subsystem, RDMA Driver Subsystem
Development	C, python, tmux, vim

REFEREES

- Antonio J. Peña [antonio.pena@bsc.es], Manager AccelCom, Barcelona Supercomputing Center.
- Asim Usman [asim@dreambigsemi.com] Director Software Engineering, Arm (previously DreamBig Semiconductor).
- Tassadaq Hussain [tassadaq@ucerd.com], Associate Professor, Namal University.